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RIKIN PAREKH

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PROFILE

Embedded systems and robotics engineer skilled in digital systems, microcontrollers, and low-level programming. Built processor-level architectures at IIT Gandhinagar and end-to-end IoT systems using STM32, ESP, and Raspberry Pi. Recipient of ₹3.2L+ in government grants and national hackathon finalist.

EDUCATION

Bachelor of Technology in Information and Communication Technology,
Pandit Deendayal Energy University — Expected 2027

- **CGPA:** 9.08/10.0 [upto 5th Semester]
- **Coursework:** Embedded Systems, Digital Signal Processing and Communication, AI Systems, Data Structures and Algorithm, Computer Organisation and Design, 3D Modeling

HSC - 12th Grade, Best Higher Secondary School — 2023

EXPERIENCE

Electronics and Embedded Systems Engineer Intern,
Center for Creative Learning, IIT Gandhinagar — July 2025 – February 2026

- Designed advanced digital-systems prototypes under Prof. Manish Jain for the India Semiconductor Mission, including a fully functional 8-bit breadboard computer; wrote low-level AVR assembly and custom drivers for ATmega328 and display modules.
- Demonstrated projects to national dignitaries including Hon. Shri S. Somnath, Union Ministers Pradhan & Vaishnaw, and senior officials from NCERT, CBSE, NIC, and MoHA.

Embedded Systems Engineer Intern, STEMbotix — May 2025 – June 2025

- Built a production-ready quadruped robot and an interactive STEM game system, developing firmware, peripheral interfaces, and hardware integration across STM32 and other microcontroller platforms.

SKILLS

Programming Languages	Python, C, Embedded C, Assembly, MATLAB
Microcontrollers and Platforms	AVR (Arduino), ESP32/ESP8266, ARM Cortex-M (STM32), Raspberry Pi
Frameworks and Tools	Microchip Studio, OpenCV, Django, Flask, Rest APIs, Solid Works, Datamatics TruBot, Git, Numpy, Pandas, Linux

PROJECTS

Print&Go – Smart Printing Kiosk — 2025 - Present
Received ₹2.5L grant under the Government of Gujarat's Student Startup & Innovation Policy (SSIP).

- Designed a 24×7 self-service document-printing vending machine with peer-to-peer encryption, secure file upload, and automated payment/printing workflow.

SegReClaim – Incentive-Based Waste Collection System — 2025 - Present
Received ₹70k grant under the Government of Gujarat's Student Startup & Innovation Policy (SSIP).

- Built a reverse-vending IoT system that automatically identifies, segregates, and rewards recyclable waste deposits, promoting sustainable disposal practices.

Underwater ROV 2024

Cleared Level I of Robofest and was awarded ₹50,000 funding for continued development.

- Developed ideation and a prototype for a remotely operated underwater vehicle (ROV) for submerged structural inspection, engineering a BLDC motor-driven propulsion system controlled via Raspberry Pi for real-time 6-DOF maneuvering; secured ₹50,000 in development funding upon advancing at Robofest, validating the design's technical and commercial potential.

8 bit CPU on breadboard from scratch 2025

- **Engineered a fully functional 8-bit CPU from scratch on a breadboard**, implementing a custom programmable control unit using EEPROM as a lookup-based instruction decoder, a Python-scripted binary ROM generator, and an 8×8-bit RAM architecture using 74LS189 ICs — demonstrating deep understanding of digital logic and computer architecture at the hardware level.
- **Designed and executed a complete instruction execution pipeline** featuring a 8-bit program counter (with 74LS161) with JMP and branch support, a variable-frequency clock module (555 timer) with manual/auto switching, and successfully validated the system by running the Fibonacci series — proving end-to-end functionality from clock signal to program output.

ACHIEVEMENTS

- **Top 5 Finalist – Nirman 1.0, PDEU SOM & IIC** Apr, 2025
  Secured 5th position in a university-level innovation and entrepreneurship competition. Received a cash prize of ₹8000. Project details are confidential due to ongoing development potential.
- **Two Time Finalist - Datamatics Hackathon, IIT Bombay** Dec, 2024
  Dec, 2023
  Developed an RPA solution to automate business processes, enhancing efficiency and accuracy.
- **Winner - Round 1, Robofest (Underwater ROV Category)** Aug, 2024
  Led a team in designing and prototyping a remotely operated underwater vehicle for structural analysis of submerged infrastructure. Secured ₹50,000 in funding.

CO-CURRICULAR EXPERIENCE

- **Cretus Robotics Club - Technical Lead**
  Engaged in hands-on robotics workshops, hackathons, and technical competitions, expanding expertise in embedded systems, automation, and AI.
- **Atal Tinkering Lab - Mentor & Workshop Conductor**
  Supervised and organized STEM workshops for students, promoting innovation in robotics and IoT.

CERTIFICATIONS & LEARNING

- **Silver Elite Certification – Drone Systems & Control (IISc Bangalore, Score: 83%)** 2025
- **IBM SkillsBuild: Building AI Solutions Using Advanced Algorithms and Open Source Frameworks** 2025
- **Hosted Robotics & Embedded Systems Workshop at New LJ university** 2026